SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA14 **COURSE NAME: Compiler Design**

COURSE OUTCOME COVERED

CO4: *Examine intermediate code generation techniques to enhance code optimization (BL4)*

Assessment Tool Weightage: 40%

Duration :60 Mins
On Class
QUESTIONS: 5 **Total Marks:50**

Annexure A

CLASS TEST

Code of Conduct:

I Arunprasath R 192524077(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
ARUNPRASATH R

Q. No	Understanding of Concepts (25)	Explanation (30)	Application (20)	Organization(15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	**/ 30**	**/ 20**	**/ 15**	**/ 10**	**/ 100**

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CO4 AT1

CLASS TEST

Q1. Write down the translation scheme to generate code for assignment statement. List the scheme for generating three address code for the assignment statement $x = (a + b) * (c + d)$

Q2.Consider the following Boolean expression: (a > b) && (c < d) || (e < f)

The grammar for Boolean expressions is given as:
$B \rightarrow B1 \parallel B2$, $B \rightarrow B1$ && $B2$, $B \rightarrow !\,B1$, $B \rightarrow E1$ relop $E2$, $B \rightarrow$ true , $B \rightarrow$ false, $E \rightarrow$ id

(i). Explain how operator precedence and associativity affect the evaluation of the given Boolean expression.

(ii). Construct the Three Address Code (TAC) for the given Boolean expression using short-circuit evaluation.

(iii). Represent the control flow of the Boolean expression using appropriate labels and conditional jump statements.

Q3.Given the declaration sequence

int a, b[10];

float c, d; derive the intermediate representation including symbol table entries, type sizes, and offset calculations.

Q4.Consider the following program fragments:

```
(i) . while (A < C && B > D) do
   if (A == 1) then
      C = C + 1
   else
      while (A <= D) do
         A = A + B
```

Generate the Three Address Code (TAC) for the above program fragment. Clearly show: (a) Conditional and unconditional jump statements, (b) Use of labels for control flow, (c) Evaluation of Boolean expressions

```
(ii). main()
{
  int i;
  int a[10];
  i = 1;
  while (i <= 10)
  {
    a[i] = 0;
    i = i + 1;
} }
```

Translate the executable statements of the above C program into Three Address Code (TAC). Clearly indicate: (a) Representation of array elements, (b) Loop control using labels, (c) Increment and assignment operations

Q5. Use backpatching to generate intermediate code for:

```
if (a < b && c < d)
  x = a + b;
else
  x = c + d;
```

Show the intermediate instructions before and after backpatching. (10 Marks)

COA AT1 - class Test

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1) The syntax for the translation scheme to generate code for assignment statement is

$S \rightarrow id = E$
$E \rightarrow E_1 + E_2$
$E \rightarrow E_1 * E_2$
$E \rightarrow (E)$
$E \rightarrow id$

$S \rightarrow id = E$
{ emit(id. lexeme = E. place); }

$E \rightarrow E_1 + E_2$
{ E. place = newtemp();
emit (E. place = E_1. place + E_2.place); }

$E \rightarrow E_1 * E_2$
{ E.place = newtemp();
emit (E.place = E_1. place * E_2. place); }

$E \rightarrow (E)$
{ E.place = E_1. place ; }

$E \rightarrow id$
{ E.place = id. lexeme ; }.

The scheme for generating three address code for the assignment statement

$x = (a+b) * (c+d)$

$t_1 = a+b$
$t_2 = c+d$
$t_3 = t_1 * t_2$.
$x = t_3$.

5)

if (a<b && c<d)
x = a+b
else
x = c+d

100 : if a<b goto 102
101 : goto 107
102 : if c<d goto 104
103 : goto 107
104 : $t_1 = a+b$
105 : $x = t_1$
106 : goto 109
107 : $t_2 = c+d$
108 : $x = t_2$
109 :

backpat cn (100, 102.
backpat (102, 104
backpat (103, 107
backpat (106, 107.

4) (i)

100 : if A<C then goto 102
101 : goto 115
102 : if B>D then goto 104.
103 : goto 115
104 : if A=1 then goto 106
105 : goto 109
106 : $t_1 := C+1$
107 : $C := t_1$
108 : goto 100
109 : if A<=D then goto 111
110 : goto 100
111 : ~~goto 11~~ $t_2 := A+B$
112 : $A := t_2$
113 : goto 109
114 : goto 100
115 : *

ii) a) conditional & unconditional:-
if i <= 10 then goto 103

b) loop control using labels:-
L1 & L2 L1 = True
L2 = False.

c) Increment assignment op:
i = i+1 increment

1)

100 : $i := 1$
101 : if $i <= 10$ then goto <u>103</u>
102 : goto <u>108</u>
103 : $t_1 := i$
104 : $t_2 := 4 * t_1$
105 : $a[t_2] := 0$
106 : $i := i+1$
107 : goto <u>101</u>
108 : stop.

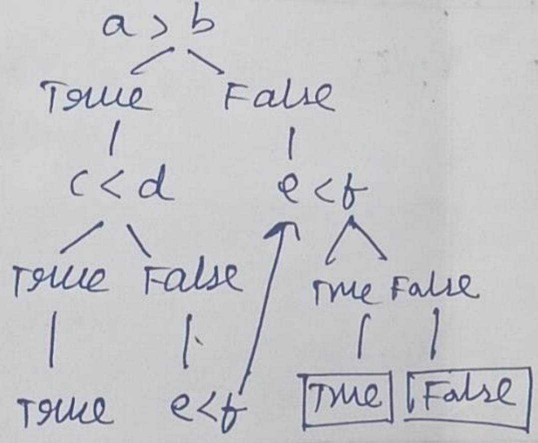
2)

(i) operator procedure:

&&
||
>
<

((a>b) && (c<d)) || (e<f)

ii)



(iii) <u>TAC</u>

1. if a>b got L1
2. goto L2
3. L1 :- if c<d goto L3
4. goto L2
5. L2: if e<f goto L3
6. goto L4
7. L3: True

8 : goto (10)
9 : L4:-
False
10 : -

(3) int a, b[10];
float c, d;

int = 4 bytes
float = 4 bytes.

a	= int	4	bytes
b	= int [10]	40	bytes
c	= float	4	bytes
d	= float	4	bytes
	Total =	52	bytes

Intermediate representⁿ

a = size = 4, offset = 0
b = size = 40, offset = 4
c = size = 4, offset = 44
d = size = 4, offset = 48

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps